

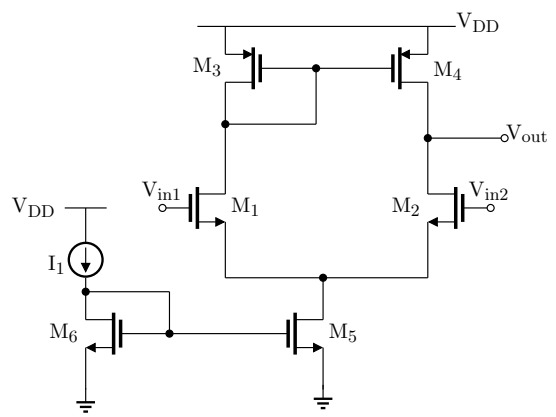
Name of the student: _____ ID. No. _____

Average: 4.24/15

Question	Marks	Score
1	9	
2	6	
Total:	15	

Question 1

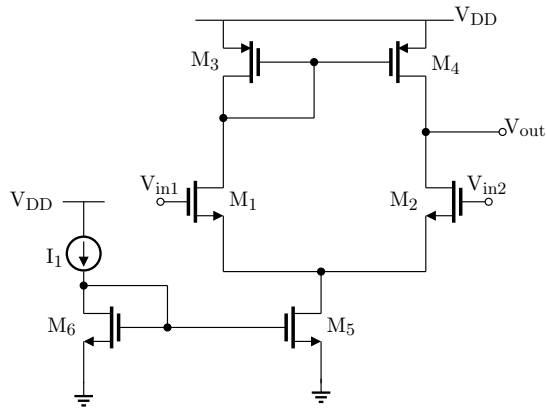
For the circuits shown below, $(W/L)_{1,2,5}=15\mu\text{m}/1\mu\text{m}$, $(W/L)_{3,4}=30\mu\text{m}/2\mu\text{m}$ and $(W/L)_6=7.5\mu\text{m}/1\mu\text{m}$ and $I_1=20\text{ }\mu\text{A}$ (ideal current sources). Assume $\lambda_n=0.1\text{ V}^{-1}$ for $L=1\text{ }\mu\text{m}$, $\lambda_p=0.2\text{ V}^{-1}$ for $L=1\text{ }\mu\text{m}$, $\mu_n C_{ox}=100\text{ }\mu\text{A}/\text{V}^2$, $\mu_p C_{ox}=50\text{ }\mu\text{A}/\text{V}^2$, $V_{th,n}=0.4\text{ V}$, $|V_{th,p}|=0.5\text{ V}$ and $V_{DD}=1.8\text{ V}$. Neglect body effect. [9]



Gain of the circuit	61.25 V/V
Output voltage swing	1.18 V
$V_{in,CM,Max}$	1.47 V
$V_{in,CM,Min}$	0.79 V
Power Dissipation	108 μW

Question 2

For the circuits shown below, $(W/L)_{1,2,5}=15\mu\text{m}/1\mu\text{m}$, $(W/L)_{3,4}=30\mu\text{m}/2\mu\text{m}$ and $(W/L)_6=7.5\mu\text{m}/1\mu\text{m}$. For Load capacitor of 2 pF, the slew rate is 10 V/ μs . Assume $\lambda_n=0.1\text{ V}^{-1}$ for $L=1\mu\text{m}$, $\lambda_p=0.2\text{ V}^{-1}$ for $L=1\mu\text{m}$, $\mu_n C_{ox}=100\mu\text{A}/\text{V}^2$, $\mu_p C_{ox}=50\mu\text{A}/\text{V}^2$, $V_{th,n}=0.4\text{ V}$, $|V_{th,p}|=0.5\text{ V}$ and $V_{DD}=1.8\text{ V}$. Neglect body effect. [6]



Value of I_1	10 μA
Bandwidth of amplifier in Hz	159.15 KHz
Gain Bandwidth Product in Hz	13.77 MHz
Power Dissipation	54 μW